

Security Incident Report

SSH Brute Force Detection

INCIDENT CLASSIFICATION

Credential Access — Brute Force

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1 - Executive Summary

Following the previously identified reconnaissance activity, repeated SSH authentication failures were detected against the Windows host (10.0.0.30). The failed logon attempts were generated by a brute-force attack targeting the account testuser@SOC.local. Wazuh identified multiple Windows Security Event ID 4625 records indicating invalid credentials and classified the activity as repeated authentication failures. The incident represents a Credential Access attempt and is mapped to MITRE ATT&CK T1110; Brute Force.

2 - Timeline

Time	Event	Source
2026-06-23 22:08:39	SSH brute-force attack initiated (Hydra)	Hydra
2026-06-23 22:08:45	Brute-force attempts completed (0 valid password found)	Hydra
2026-06-23T22:08:46.345	Windows generated Event ID 4625 (failed logon) and alert ingested by Wazuh	Wazuh SIEM

3 - Detection Details

Field	Value
Detection Source	Windows Security Log
SIEM	Wazuh
Event ID	4625
Rule ID	60122
Rule Level	5
Account	testuser@SOC.local
Process	sshd.exe
Logon Type	8
MITRE ATT&CK	T1110 – Brute Force

4 - Evidence Collected

Figure 1 — Hydra Brute Force Attack

```
(kali# kali) ~ (/Desktop)
$ hydra -l testuser -P /home/kali/Desktop/passw.txt -u ssh://10.0.0.30
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-bin
ding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-06-23 22:08:39
(WARNING) Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 16 tasks per 1 server, overall 16 tasks, 60 login tries (1:1/p:60), ~5 tries per task
[DATA] attacking ssh://10.0.0.30:22/
[VERBOSE] Resolving addresses ... [VERBOSE] resolving done
[INFO] Testing if password authentication is supported by ssh://testuser@10.0.0.30:22
[INFO] Successful, password authentication is supported by ssh://10.0.0.30:22
[ERROR] could not connect to target port 22: Socket error: Connection reset by peer
[ERROR] ssh protocol error
[VERBOSE] Disabled child 7 because of too many errors
[STATUS] attack finished for 10.0.0.30 (waiting for children to complete tests)
1 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-06-23 22:08:45
```

Figure 2 — Wazuh Authentication Failure Alert

# data.win.eventdata.workstationName	CLIENT
# data.win.system.channel	Security
# data.win.system.computer	Client.SOC.local
# data.win.system.eventID	4625
# data.win.system.eventRecordID	88859
# data.win.system.keywords	0x8010000000000000
# data.win.system.level	0
# data.win.system.message	> "An account failed to log on. Subject: Security ID: S-1-5-18 Account Name: CLIENTS Account Domain: SOC

Figure 3 — Alert Technical Details

# _index	wazuh-alerts-4.x-2026.06.23
# agent.id	002
# agent.ip	10.0.0.30
# agent.name	Amin
# data.win.eventdata.authenticationPackageName	Negotiate
# data.win.eventdata.failureReason	%2313
# data.win.eventdata.keyLength	0
# data.win.eventdata.logonProcessName	Advapi
# data.win.eventdata.logonType	8
# data.win.eventdata.processId	0xb8c
# data.win.eventdata.processName	C:\Windows\System32\OpenSSH\sshd.exe
# timestamp	Jun 23, 2026 @ 22:08:46.345
# rule.id	60122
# rule.level	5

5 - Investigation Findings

- The attack occurred shortly after the reconnaissance activity documented in Incident Report #1, indicating a possible continuation of the attack lifecycle. However, the source IP address of the brute-force attempt was not directly captured in the available evidence (Hydra output and Windows Event ID 4625 logs); the association with the previously identified host (10.0.0.99) is based on timing proximity and attack-chain logic rather than confirmed log correlation.
- Multiple failed SSH authentication attempts were observed.
- All attempts targeted the account testuser@SOC.local
- The authentication requests were processed by OpenSSH (sshd.exe).
- No successful authentication associated with this activity was confirmed during this incident.

6 - Impact Assessment

Category	Assessment
Confidentiality	No impact
Integrity	No impact
Availability	No impact

- **Note:** While no credentials were compromised, repeated failed authentication attempts pose a risk of account lockout (Availability) or eventual success if weak/default credentials exist.

7 - Root Cause Analysis

The attacker attempted to gain unauthorized access by repeatedly submitting invalid credentials to the SSH service exposed on the target host. The SSH service had previously been identified during the reconnaissance phase documented in Incident Report #1. While the source of this brute-force activity could not be conclusively linked to the same IP address (10.0.0.99) due to the absence of source-IP data in the captured logs, the short time interval between the two incidents strongly suggests a continuation of the same attack lifecycle. The observed activity is consistent with a password guessing or brute-force attack against a valid user account.

8 - Recommendations

- Enforce strong password requirements.
- Restrict SSH access to trusted IP ranges or VPN, and disable the service when not actively required.
- Monitor repeated authentication failures.
- Correlate authentication events with previous reconnaissance activity.
- Capture and correlate source IP addresses in SSH/application-level logs to confirm attack-chain continuity between reconnaissance and credential-access phases

9 - Conclusion

The incident was confirmed as a brute-force authentication attempt targeting the SSH service on the Windows host. Multiple failed logon events (Event ID 4625) were successfully detected and correlated by Wazuh. Based on the investigation, the activity is assessed as the logical continuation of the reconnaissance phase documented in Incident Report #1 and is mapped to MITRE ATT&CK T1110; Brute Force. Although no successful compromise was confirmed during this incident, the repeated authentication attempts demonstrate an active effort to obtain valid credentials and justify continued monitoring for subsequent attack stages.